LabelGuard AI — Best Model for Each Task

Team Discussion Reference | SIH 26034 | September 2026

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1. OCR — Reading Text from Package Labels

Model: PaddleOCR v4 (Baidu, Apache 2.0, Free)

How it works: Uses a 3-stage pipeline — DBNet++ detects text regions, a classifier corrects text orientation, and SVTR-LCNet recognizes characters. Supports 80+ languages including Hindi, Tamil, Telugu, and Bengali out of the box.

Workflow:

```
Package Image → DBNet++ (detect text boxes) → Orientation
```

- Size: ~15MB (lite model for mobile)
- Runs offline: Yes
- Indian languages: 12+

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2. Layout Analysis — Identifying Label Sections

Model: DocLayout-YOLO v1.0 (OPENDATALAB, 2024, Apache 2.0)

How it works: A YOLO variant re-engineered for document layouts. It detects structured regions like tables, text blocks, headers, and stamps instead of generic objects. Trained on 11 diverse document datasets, it segments a product label into meaningful zones in one forward pass.

Workflow:

```
Package Image → DocLayout-YOLO → Bounding boxes for each
  [MRP zone, Ingredients list, Manufacturer block, Nutrit
```

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3. Structured Field Extraction — Image to JSON (OCR-Free)

Model: Donut v1 — Document Understanding Transformer (Naver/ Clova, MIT License)

How it works: An OCR-free end-to-end model. It takes a raw label image as input and directly outputs structured JSON fields — no separate OCR step needed. Uses a Swin Transformer vision encoder paired with a BART text decoder to generate key-value pairs.

Workflow:

```
Package Image → Swin Vision Encoder → BART Text Decoder →
{
  "mrp": "₹45",
  "net_qty": "500g",
  "mfg_date": "07/2026",
  "expiry": "01/2027",
  "fssai": "10012345678901",
  "manufacturer": "Parle Products, Mumbai 400057"
}
```

- Best for: Fast extraction directly from images
- Limitation: Weaker on code-mixed Hindi-English text (use

IndicBERT for that)

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4. Indian Language NER — Deep Text Understanding After OCR

Model: IndicBERT v2 (AI4Bharat, IISc Bangalore, 2023, MIT License)

How it works: A lightweight BERT model pre-trained on 24 Indian languages using IndicCorp v2 (27 billion tokens). Only 33M parameters — runs on CPU easily. Fine-tune it for Named Entity Recognition (NER) to tag OCR text as MRP, DATE, ADDRESS, FSSAI_NUMBER, PRODUCT_NAME, etc. Handles Hindi-English code-mixed text better than any other model.

Workflow:

```
PaddleOCR extracted text:
  "Parle-G Gold Glucose Biscuits Net Wt 200g MRP ₹30 incl
   Mfg by Parle Products Pvt Ltd Vile Parle Mumbai 400057

→ IndicBERT v2 NER →

  "Parle-G Gold"              → BRAND_NAME
  "Glucose Biscuits"          → GENERIC_NAME ✓
  "200g"                      → NET_QUANTITY ✓
  "₹30"                       → MRP ✓
  "inclusive of all taxes"     → MRP_QUALIFIER ✓
  "Parle Products Pvt Ltd"    → MANUFACTURER ✓
  "Vile Parle Mumbai 400057"  → ADDRESS ✓ (has city + PIN
  "10012345678901"            → FSSAI_NUMBER ✓
```

- Size: 33M params (very lightweight)
- Best for: Code-mixed Indian text, brand vs generic name classification, address completeness
- Runs on: CPU (no GPU needed)

How IndicBERT and Donut work together:

Step	Model	Job
Image → Fields (fast)	Donut v1	Extracts fields directly from image
OCR Text → Deep Understanding	IndicBERT v2	Validates, classifies, handles Indian language nuances
Merge both	Rule Engine	Best of both for final compliance check

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5. Symbol and Logo Detection — Veg/Non-Veg, FSSAI Logo, ISI Mark

Model: RT-DETR v2 — Real-Time Detection Transformer (Baidu, 2024, Apache 2.0)

How it works: First real-time Vision Transformer detector. Uses self-attention to understand the entire label at once, so it handles tightly packed symbols (veg dot next to FSSAI logo next to barcode) without confusion. No anchor boxes or NMS post-processing.

Workflow:

```
Package Image → Hybrid CNN-Transformer Backbone → Transfo
  Detected symbols with bounding boxes + confidence:
    [Veg Symbol: 0.97, FSSAI Logo: 0.94, Barcode: 0.99, R
```

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6. Font Size Measurement — Rule 7 Character Height Compliance

Model: DocTR v0.9 (Mindee, 2024, Apache 2.0)

How it works: A document text recognition library that outputs precise per-word and per-character bounding boxes with exact pixel coordinates. Measure character height from the bounding box, then convert pixels to millimetres using a known reference dimension on the package.

Workflow:

```
Package Image → DocTR text detection → Per-character boun
→ Height in pixels → Convert using reference scale
→ Compare against Rule 7 minimum (e.g., 2.5mm for 100-500
→ PASS / FAIL
```

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7. Tampering and Counterfeit Detection — MRP Overprinting

Model: PatchCore v1.0 (Amazon, Apache 2.0)

How it works: A self-supervised anomaly detector trained ONLY on genuine product images. It builds a memory bank of normal visual features. At inference, any deviation — sticker overlay, reprint, font mismatch, color shift — gets flagged as an anomaly without ever needing fake/tampered training data.

Workflow:

```
Training: 50-100 genuine product images → WideResNet-50 f

Inference: New package image → Extract patches → Compare
→ Anomaly score per region
→ "Potential tampering detected near MRP area" (score: 0.
```

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8. Image Quality Check — Blur, Lighting, Completeness

Model: TOPIQ v0.1 (IQA-PyTorch, 2024, BSD License)

How it works: A no-reference image quality assessment model that predicts a quality score (0 to 1). Uses cross-scale attention to evaluate sharpness, noise, and exposure at multiple resolutions simultaneously. Runs in real-time on mobile.

Workflow:

```
Camera preview frame → TOPIQ model → Quality score (0–1)
  → Score > 0.7:   "Good quality, proceed with scan" ✓
  → Score 0.4–0.7: "Hold steady / Move closer"
  → Score < 0.4:   "Too blurry / Improve lighting"
```

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9. LLM for Legal Reasoning — Edge Cases and Explanations

Model: Gemma 3 27B (Google, 2025, Gemma License, Free)

How it works: Google's latest open-weight LLM with strong reasoning. Use with RAG — chunk the 3 legal PDFs into ChromaDB, retrieve relevant sections for edge cases, and ask Gemma to reason and explain in plain language with rule citations.

Workflow:

```
Edge case: "Label says 'Net Wt. Approx 500g' — is 'approx
→ RAG retrieves: Rule 6(1)(b) from Legal Metrology Rules
→ Gemma 3 reasons: "No. Rule 6(1)(b) requires a definite
   'Approximately' makes it indefinite — this violates th
```

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10. Multimodal Vision-Language — Direct Image-to-Analysis

Model: Qwen 2.5 VL 7B (Alibaba, 2024, Apache 2.0)

How it works: A Vision-Language model that processes images at native resolution and reads dense text directly. Feed it a label photo with a compliance prompt and it reads, extracts, and reasons in one pass. 7B params, runs on a single GPU.

Workflow:

```
Package Image + Prompt: "Check this label for Legal Metro
→ Qwen 2.5 VL processes image natively
→ "MRP: ₹45 (present ✓), FSSAI: not detected (MISSING ✗),
   Expiry: Jan 2027 (valid ✓), Address: incomplete — no P
```

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11. GIS / Mapping — Inspection Locations and Violation Heatmaps

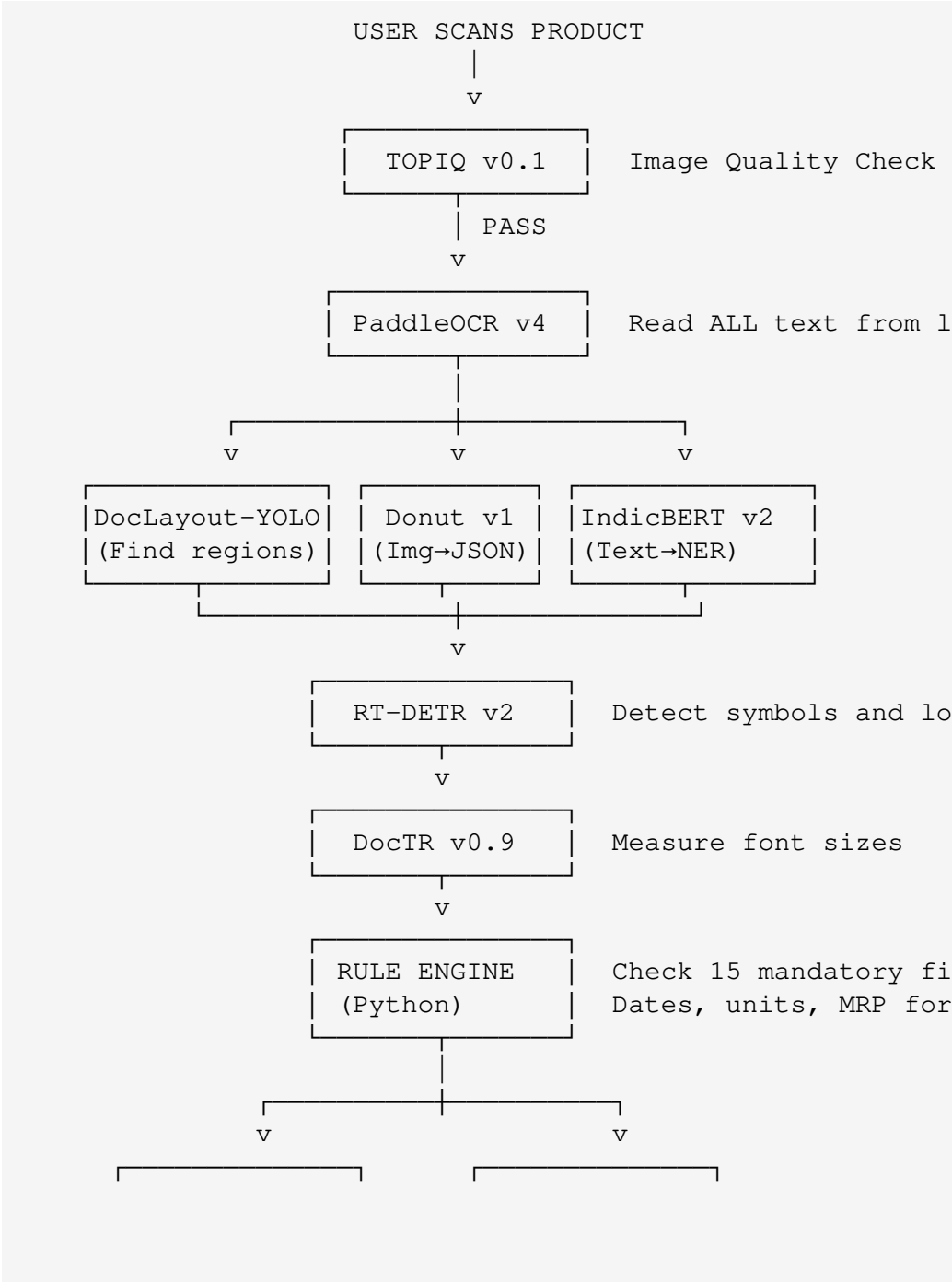
Tool: Leaflet.js v1.9 + OpenStreetMap (BSD-2 License, Free)

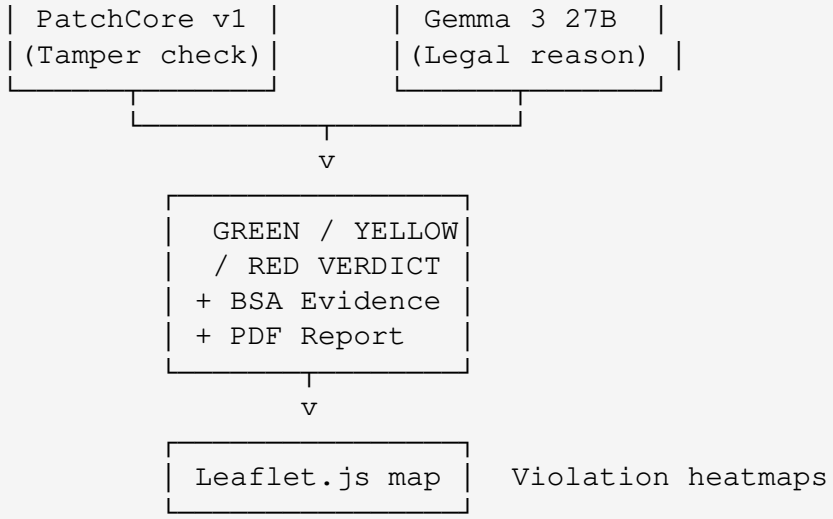
How it works: Lightweight JavaScript mapping library using free OpenStreetMap tiles. Plot inspection pins, violation clusters, and heatmap layers. Zero cost, no API key required.

Workflow:

```
Inspection data (lat, lng, violation_type, date) → Leafle
→ Pin markers + Heatmap overlay (red = high violations, g
→ Click pin → popup with inspection details and report li
```

Full Pipeline — How All Models Connect





Summary Table

#	Task	Model	Params	License	Offline	GPU
1	OCR	PaddleOCR v4	~15MB	Apache 2.0	Yes	No
2	Layout	DocLayout-YOLO v1.0	~25M	Apache 2.0	Yes	Preferred
3	Field Extraction	Donut v1	~200M	MIT	Yes	Yes
4	Indian NER	IndicBERT v2	33M	MIT	Yes	No (CPU)
5	Symbol Detection	RT-DETR v2	~50M	Apache 2.0	Yes	Preferred
6	Font Measurement	DocTR v0.9	~30M	Apache 2.0	Yes	No

7	Tamper Detection	PatchCore v1.0	~25M	Apache 2.0	Yes	Preferred
8	Image Quality	TOPIQ v0.1	~10M	BSD	Yes	No
9	Legal LLM	Gemma 3 27B	27B	Gemma	Local	Yes
10	Vision-Language	Qwen 2.5 VL 7B	7B	Apache 2.0	Local	Yes
11	GIS Mapping	Leaflet.js v1.9	N/A	BSD-2	No	No

Team Decision Points

Choice 1: Field Extraction Strategy

- Option A: Donut only (faster, simpler, weaker on Indian languages)
- Option B: PaddleOCR + IndicBERT only (better Indian text, needs OCR step)
- Option C: Both Donut + IndicBERT (best accuracy, more complex)

Choice 2: LLM Deployment

- Option A: Gemma 3 27B locally (needs 16GB+ GPU, fully offline)
- Option B: Gemini API cloud (no GPU needed, needs internet, free tier)
- Option C: Phi-4 14B locally (smaller, runs on 8GB GPU)

Choice 3: Vision-Language Model

- Option A: Qwen 2.5 VL as primary scanner (simpler, one model)
- Option B: Full multi-model pipeline (more accurate, more complex)
- Option C: Qwen 2.5 VL as fallback when main pipeline is uncertain

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All models are free and open-source. No paid API required. Prepared for SIH 26034 Team Discussion — September 2026